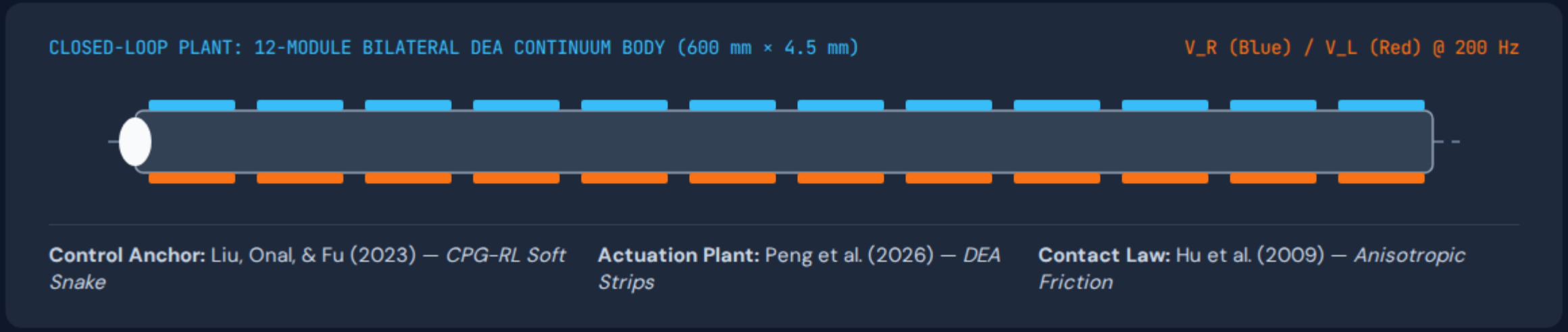
ADVANCED CONTROL SYSTEMS · MIDTERM PROGRESS REVIEW

Hierarchical CPG-Regulated RL for a Dielectric-Elastomer Soft Snake Robot



Control Anchor: Liu, Onal, & Fu (2023) — *CPG-RL Soft Snake*

Actuation Plant: Peng et al. (2026) — *DEA Strips*

Contact Law: Hu et al. (2009) — *Anisotropic Friction*

BITS Pilani, Hyderabad Campus · October 2026

70% Control & Learning · 30% Plant & Physics

Why Standard Model-Free RL Fails on Soft Robots

-80%

Sim-to-Real Locomotion Speed Collapse

End-to-end PPO exploits simulator dynamics with ~10 Hz chattering commands.

Continuum Underactuation

Infinite passive DOFs with viscoelastic phase lag.

High-Frequency Chattering

Soft actuators low-pass filter 10 Hz RL noise.

Hierarchical Cascade Solution

Option-Critic RL brain drives Matsuoka CPG spine.

Four-Layer Cascaded Control Hierarchy

L4 · BRAIN — **Proximal Policy Option-Critic (PPOC) / Classical ILOS**
Outputs tonic bias u_i (200 Hz) & frequency option K_f

L3 · SPINE — **12-Module Coupled Matsuoka CPG Network**
Nonlinear ODE filter → Smooth ~1.6 Hz curvature ref $\kappa_{ref}(t)$

L2 · INNER LOOP — **Capacitive Self-Sensing PI Curvature Regulator**
200 Hz ΔC_i feedback → Bilateral strip voltages V_L, $V_R \leq 450$ V

L1 · PLANT — **DEA Silicone Continuum + Anisotropic Coulomb Floor**
Maxwell stress $p = \varepsilon(V/h)^2$ · Friction ratio $\eta = \mu_n / \mu_t$

Solid-State Actuation, Self-Sensing & Contact Mechanics

Electro-Elastic Transfer Law

$$p_i = \epsilon_0 \epsilon_r \frac{V_i^2}{h} \Rightarrow \kappa_i = GV_i^2$$

Bilateral Drive: 24 strips across 12 modules at 450 V (54 kPa).

Peak Authority: Achieves 13.6 m^{-1} curvature with 20% pull-in margin.

Collocated Observability: Capacitance C_i senses curvature at 200 Hz.

1. Voltage Input	450 V across 10 µm dielectric films
2. Maxwell Stress	Quadratic electrostatic pressure (54 kPa)
3. Module Curvature	Inextensible neutral axis bends to 13.6 m^{-1}
4. Anisotropic Contact	Coulomb friction ratio $\eta = \mu_n / \mu_t > 1.7$

Coupled Matsuoka CPG & Steering Linearization

Half-Center Oscillator Dynamics

$$K_f \tau_r \dot{x}_{i,e} = -x_{i,e} - bv_{i,e} - wy_{i,f} + u_i^e$$

Dynamical Low-Pass Filter: Converts discrete RL inputs into smooth ~1.6 Hz limit cycles.

Phase-Locked Chain: Inter-module synapses enforce traveling wave propagation.

Describing Function Proof

$$u_i^f = 1 - u_i^e \Rightarrow \Delta\kappa_i \propto 2u_i^e - 1$$

Complementary Tonic Drive: Binds extensor and flexor inputs symmetrically.

Linear Steering Manifold: Output DC curvature bias scales strictly linearly with u_i.

Multi-Rate Option-Critic (PPOC) & FOC Protection

Control Channel	Update Rate	Control Variable	Physical Role
Intra-Option Policy	200 Hz (Fast)	Tonic Steering Bias u_i	Continuous curvature offset for heading alignment
Macro-Option Policy	Multi-Step (Slow)	Time-Constant Gain K_f	Locks wave frequency until termination β(s)

Free-Response Oscillation Constraint (FOC): c = 0.75

Bounds tonic steering inputs to $u_i \in [0.25, 0.75]$ so RL policies never quench CPG wave amplitude.

c = 0.75

Classical Baseline: PI Curvature Loop + ILOS Guidance

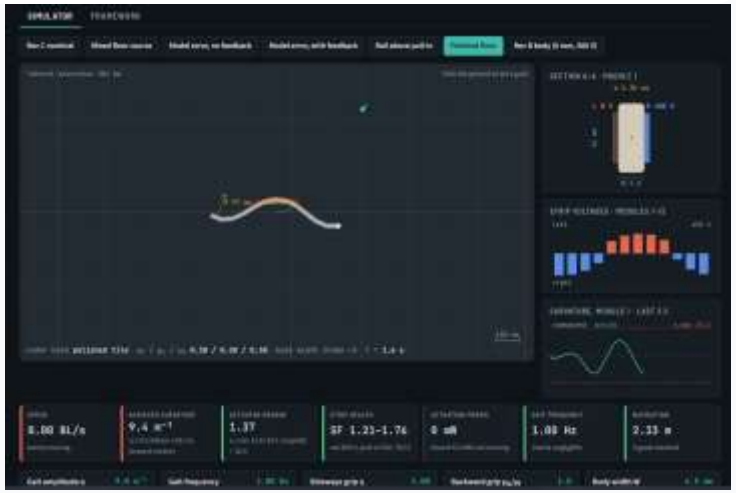
465 mm
Advance per cycle on sand ($\eta = 3$)

0.78 BL/s
Nominal speed at 1.0 Hz wave

0 mm/s
Stalls on isotropic tile ($\eta = 1.0$)







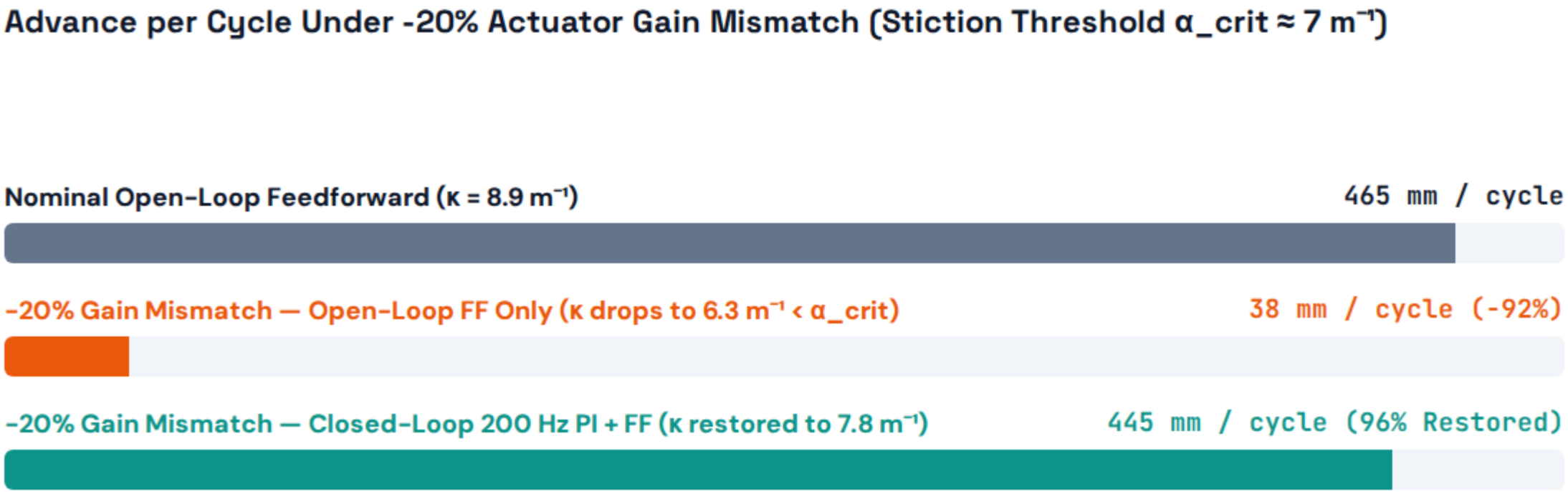
Nominal Waypoint Tracking (Sand)

ILOS steering bias smoothly aligns heading to target ring under uniform anisotropic grip.

Unmodeled Friction Failure (Tile Patch)

Fixed-wavelength classical gait cannot adapt frequency or posture when lateral grip drops.

Curvature Thresholds & 200 Hz PI Disturbance Rejection



Coulomb Bifurcation: Reverses direction if η < 1.7.

Inertial Cutoff: Flat response to 1.5 Hz matches 1.6 Hz CPG.

Input Saturation, Slew-Rate Governor & Slip Limits

+36% Margin

Steering Authority Headroom

4.5 mm body width yields 13.6 m^{-1} max curvature above 10 m^{-1} wave demand.

0 / 24 Failed

Slew-Rate Reference Governor

45 V/µm voltage rail plus soft-start ramp eliminates dielectric breakdown.

L/2 Stall Limit

Spatial Disturbance Boundary

Classical gait crosses 150 mm (L/4) polished band; stalls at 300 mm (L/2).

Closing the Sim-to-Real Gap: 3D Continuum FEM in SOFA

1D Python Model Limitation

Kinematically imposes curvature $\kappa(s,t)$; ignores floor reaction torque against strips.

Two-Way Continuum Coupling in SOFA

Hexahedral silicone FEM mesh matches beam theory within 0.3% under true contact loading.



Three-Phase Control Systems Implementation Plan

PHASE 1 · WEEKS 1–2

3D SOFA Gym Wrapper

Expose 12-module curvature states and randomized friction bands to Gymnasium API.

PHASE 2 · WEEKS 3–4

Matsuoka CPG Spine

Implement 24 coupled neural ODEs with complementary tonic inputs and FOC (c = 0.75).

PHASE 3 · WEEKS 5–6

PPOC Policy & Benchmark

Train Option-Critic agent with potential-field rewards to cross ≥ L/2 slippery bands.

Thank You · Questions & Discussion

Refs: Liu et al. (2023) · Peng et al. (2026) · Hu et al. (2009)